LAB ASSIGNMENT – 02

DATE: 19/01/2026

Name :- Pratiksha Pal

Roll No :- UG/02/BTCSEAIML/2023/064

Reg-NO: AU/2023/0010235

Section : ’E”

Q1. Make a schema first and then insert 6 documents.

- Roll_no=[1,2,3,4,5,]
- Name=["Ram","Alex","John","Bob","Mukesh",”Danny”]
- Age=[20,19,40,55,30,28]
- Salary=[546.7,333.4,666.7,678.4,245.6,546.3]
- City=["A","B","C","D","E",”F”]
- Phone_no=[123,456,122,444,567,892]

```
Office_AIML064> db.createCollection("Employees_AIML064")
{ ok: 1 }
Office_AIML064> db.Employees_AIML064.insertMany([{roll_no:1,name:"Ram",age:20,salary:546.7,city:"A",phone_no:123},{roll_no:2,name:"Alex",age:19,salary:333.4
,city:"B",phone_no:456},{roll_no:3,name:"John",age:40,salary:666.7,city:"C",phone_no:122},{roll_no:4,name:"Bob",age:55,salary:678.4,city:"D",phone_no:444},{
roll_no:5,name:"Mukesh",age:30,salary:245.6,city:"E",phone_no:567},{roll_no:6,name:"Danny",age:28,salary:546.3,city:"F",phone_no:892}])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('699b438538dc1957ec628ca0'),
    '1': ObjectId('699b438538dc1957ec628ca1'),
    '2': ObjectId('699b438538dc1957ec628ca2'),
    '3': ObjectId('699b438538dc1957ec628ca3'),
    '4': ObjectId('699b438538dc1957ec628ca4'),
    '5': ObjectId('699b438538dc1957ec628ca5')
  }
}
```

Q2. Write a query to update the name of RAM to SAM.

```
Office_AIML064> db.Employees_AIML064.updateOne({name:"Ram"},{$set:{name:"Sam"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Q3. Write a query to display only the cities present in that collection.

```
Office_AIML064> db.Employees_AIML064.find({},{"city":1,"_id":0})
[
  { city: 'A' },
  { city: 'B' },
  { city: 'C' },
  { city: 'D' },
  { city: 'E' },
  { city: 'F' }
]
```

Q4. Write a query to update the salary by 10%.

```
Office_AIML064> db.Employees_AIML064.updateMany({},{$mul:{salary:1.10}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 6,
  modifiedCount: 6,
  upsertedCount: 0
}
```

Q5. Write a query to display all the documents in ascending and descending order of age.

```
Office_AIML064> db.Employees_AIML064.find().sort({age:1})
[
  {
    _id: ObjectId('699b438538dc1957ec628ca1'),
    roll_no: 2,
    name: 'Alex',
    age: 19,
    salary: 366.74,
    city: 'B',
    phone_no: 456
  },
  {
    _id: ObjectId('699b438538dc1957ec628ca0'),
    roll_no: 1,
    name: 'Sam',
    age: 20,
    salary: 601.3700000000001,
    city: 'A',
    phone_no: 123
  },
  {
    _id: ObjectId('699b438538dc1957ec628ca5'),
    roll_no: 6,
    name: 'Danny',
    age: 28,
    salary: 600.93,
    city: 'F',
    phone_no: 892
  },
  {
    _id: ObjectId('699b438538dc1957ec628ca4'),
    roll_no: 5,
    name: 'Mukesh',
    age: 30,
    salary: 270.16,
    city: 'E',
    phone_no: 567
  },
  {
```

```
{
  _id: ObjectId('699b438538dc1957ec628ca2'),
  roll_no: 3,
  name: 'John',
  age: 40,
  salary: 733.3700000000001,
  city: 'C',
  phone_no: 122
},
{
  _id: ObjectId('699b438538dc1957ec628ca3'),
  roll_no: 4,
  name: 'Bob',
  age: 55,
  salary: 746.24,
  city: 'D',
  phone_no: 444
}
```

```
Office_AIML064> db.Employees_AIML064.find().sort({age:-1})
[
  {
    _id: ObjectId('699b438538dc1957ec628ca3'),
    roll_no: 4,
    name: 'Bob',
    age: 55,
    salary: 746.24,
    city: 'D',
    phone_no: 444
  },
  {
    _id: ObjectId('699b438538dc1957ec628ca2'),
    roll_no: 3,
    name: 'John',
    age: 40,
    salary: 733.3700000000001,
    city: 'C',
    phone_no: 122
  },
  {
    _id: ObjectId('699b438538dc1957ec628ca4'),
    roll_no: 5,
    name: 'Mukesh',
    age: 30,
    salary: 270.16,
    city: 'E',
    phone_no: 567
  },
  {
    _id: ObjectId('699b438538dc1957ec628ca5'),
    roll_no: 6,
    name: 'Danny',
    age: 28,
    salary: 600.93,
    city: 'F',
    phone_no: 892
  },
  {
    _id: ObjectId('699b438538dc1957ec628ca0'),
```

```
  _id: ObjectId('699b438538dc1957ec628ca0'),
  roll_no: 1,
  name: 'Sam',
  age: 20,
  salary: 601.3700000000001,
  city: 'A',
  phone_no: 123
},
{
  _id: ObjectId('699b438538dc1957ec628ca1'),
  roll_no: 2,
  name: 'Alex',
  age: 19,
  salary: 366.74,
  city: 'B',
  phone_no: 456
```

Q6. Write a query to display all the documents with City ❼ A,B,C.

```
Office_AIML064> db.Employees_AIML064.find({city:{$in:["A","B","C"]}})
[
  {
    _id: ObjectId('699b438538dc1957ec628ca0'),
    roll_no: 1,
    name: 'Sam',
    age: 20,
    salary: 601.3700000000001,
    city: 'A',
    phone_no: 123
  },
  {
    _id: ObjectId('699b438538dc1957ec628ca1'),
    roll_no: 2,
    name: 'Alex',
    age: 19,
    salary: 366.74,
    city: 'B',
    phone_no: 456
  },
  {
    _id: ObjectId('699b438538dc1957ec628ca2'),
    roll_no: 3,
    name: 'John',
    age: 40,
    salary: 733.3700000000001,
    city: 'C',
    phone_no: 122
  }
]
```

Q7. Write a query to display only two documents from the entire collection.

```
Office_AIML064> db.Employees_AIML064.find().limit(2)
[
  {
    _id: ObjectId('699b438538dc1957ec628ca0'),
    roll_no: 1,
    name: 'Sam',
    age: 20,
    salary: 601.3700000000001,
    city: 'A',
    phone_no: 123
  },
  {
    _id: ObjectId('699b438538dc1957ec628ca1'),
    roll_no: 2,
    name: 'Alex',
    age: 19,
    salary: 366.74,
    city: 'B',
    phone_no: 456
  }
]
```

Q8. Write a query to delete a document with ROLL_NO:5.

```
Office_AIML064> db.Employees_AIML064.deleteOne({roll_no:5})
{ acknowledged: true, deletedCount: 1 }
```

Q9. Write a query to display all the documents with AGE greater than 20.

```
Office_AIML064> db.Employees_AIML064.deleteOne({roll_no:5})
{ acknowledged: true, deletedCount: 1 }
Office_AIML064> db.Employees_AIML064.find({age:{$gt:20}})
[
  {
    _id: ObjectId('699b438538dc1957ec628ca2'),
    roll_no: 3,
    name: 'John',
    age: 40,
    salary: 733.3700000000001,
    city: 'C',
    phone_no: 122
  },
  {
    _id: ObjectId('699b438538dc1957ec628ca3'),
    roll_no: 4,
    name: 'Bob',
    age: 55,
    salary: 746.24,
    city: 'D',
    phone_no: 444
  },
  {
    _id: ObjectId('699b438538dc1957ec628ca5'),
    roll_no: 6,
    name: 'Danny',
    age: 28,
    salary: 600.93,
    city: 'F',
    phone_no: 892
  }
]
```

Q10. Write a query to display all the documents with AGE less than 20.

```
Office_AIML064> db.Employees_AIML064.find({age:{$lt:20}})
[
  {
    _id: ObjectId('699b438538dc1957ec628ca1'),
    roll_no: 2,
    name: 'Alex',
    age: 19,
    salary: 366.74,
    city: 'B',
    phone_no: 456
  }
]
```

Q11. Write a query to display all the documents with AGE equals to 20.

```
Office_AIML064> db.Employees_AIML064.find({age:{$ne:20}})
[
  {
    _id: ObjectId('699b438538dc1957ec628ca1'),
    roll_no: 2,
    name: 'Alex',
    age: 19,
    salary: 366.74,
    city: 'B',
    phone_no: 456
  },
  {
    _id: ObjectId('699b438538dc1957ec628ca2'),
    roll_no: 3,
    name: 'John',
    age: 40,
    salary: 733.3700000000001,
    city: 'C',
    phone_no: 122
  },
  {
    _id: ObjectId('699b438538dc1957ec628ca3'),
    roll_no: 4,
    name: 'Bob',
    age: 55,
    salary: 746.24,
    city: 'D',
    phone_no: 444
  },
  {
    _id: ObjectId('699b438538dc1957ec628ca5'),
    roll_no: 6,
    name: 'Danny',
    age: 28,
    salary: 600.93,
    city: 'F',
    phone_no: 892
  }
]
```

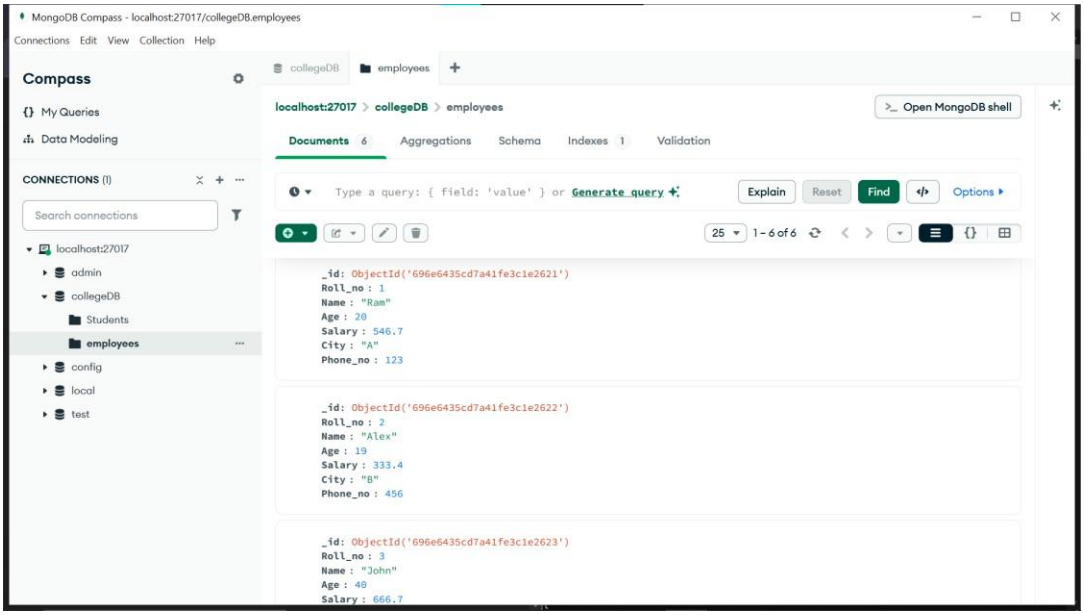
Q12. Write a query to display all the documents with AGE not equals to 20.

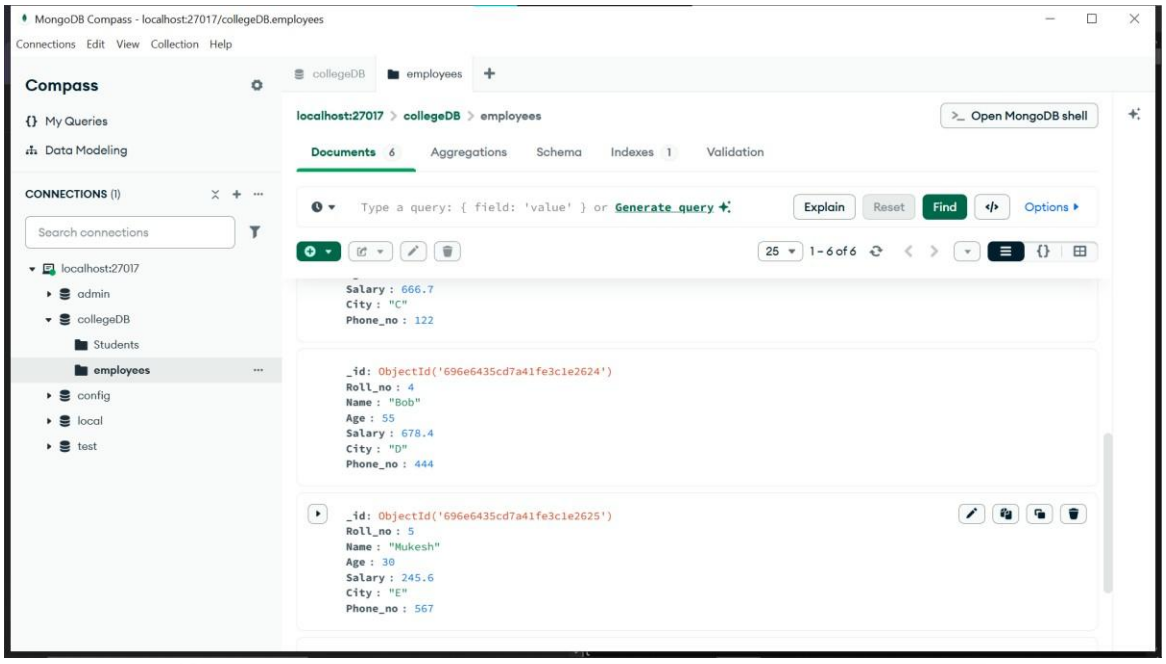
```
Office_AIML064> db.Employees_AIML064.find({age:{$ne:20}})
[
  {
    _id: ObjectId('699b438538dc1957ec628ca1'),
    roll_no: 2,
    name: 'Alex',
    age: 19,
    salary: 366.74,
    city: 'B',
    phone_no: 456
  },
  {
    _id: ObjectId('699b438538dc1957ec628ca2'),
    roll_no: 3,
    name: 'John',
    age: 40,
    salary: 733.3700000000001,
    city: 'C',
    phone_no: 122
  },
  {
    _id: ObjectId('699b438538dc1957ec628ca3'),
    roll_no: 4,
    name: 'Bob',
    age: 55,
    salary: 746.24,
    city: 'D',
    phone_no: 444
  },
  {
    _id: ObjectId('699b438538dc1957ec628ca5'),
    roll_no: 6,
    name: 'Danny',
    age: 28,
    salary: 600.93,
    city: 'F',
    phone_no: 892
  }
]
```

Q13. Write a query to display all the documents where AGE is greater than equals to 30.

```
Office_AIML064> db.Employees_AIML064.find({age:{$gte:30}})
[
  {
    _id: ObjectId('699b438538dc1957ec628ca2'),
    roll_no: 3,
    name: 'John',
    age: 40,
    salary: 733.3700000000001,
    city: 'C',
    phone_no: 122
  },
  {
    _id: ObjectId('699b438538dc1957ec628ca3'),
    roll_no: 4,
    name: 'Bob',
    age: 55,
    salary: 746.24,
    city: 'D',
    phone_no: 444
  }
]
```

MONGODB COMPASS






```
_id: ObjectId('696e6435cd7a41fe3c1e2626')
Roll_no : 6
Name : "Danny"
Age : 28
Salary : 546.3
City : "F"
Phone_no : 892
```